

ware Update Images generically to devices from many vendors.

11.20.2. Functional overview

An OTA Requestor SHALL query the OTA Provider periodically to determine the availability of new Software Images. The OTA Provider MAY learn, from backend systems inside or outside of Fabric scope, of the availability of a new Software Image for an OTA Requestor.

An OTA Requestor which has been updated using a mechanism beyond this Cluster MAY report to an OTA Provider that a Software Image update has been completed.

The OTA Provider MAY announce its presence to OTA Requestors on the Fabric to assist in discovery of this service (see [Section 11.20.7.6.1, “AnnounceOTAProvider Command”](#)).

Nodes SHALL NOT rely solely on unsolicited OTA Provider announcements to discover available OTA Providers and SHALL instead employ other means such as using OTA Provider records provisioned during Commissioning, or dynamic discovery of OTA Providers.

OTA Requestors SHALL only upgrade to numerically newer versions and OTA Providers SHALL only ever provide a numerically newer Software Image than a Node’s current version number (see [Section 11.1.5.10, “SoftwareVersion Attribute”](#)). Any functional rollback SHALL be affected by the Vendor creating a Software Image with a newer (higher) version number, but whose binary contents may match prior functionality.

All OTA Requestors SHALL support usage of a polling mechanism to send a query command to the OTA Provider in order to determine if the OTA Provider has any new Software Images for it. Polling simplifies processing for OTA Requestors that MAY need to perform special setup to get ready to receive a Software Images, such as unlocking flash or allocating space for a new Software Images.

It is ideal to have OTA Providers maintain as little state as possible since this will scale better when there are hundreds of OTA Requestors in a given Fabric. The OTA Provider is not required to keep track of what pieces of an image that a particular OTA Requestor has received.

The flow for querying the availability of a new version is done using commands of the OTA Provider Cluster. In case of a new image available matching an OTA Requestor’s request, the response to the [QueryImage](#) command SHALL contain a URI where the given image can be downloaded.

The download protocol is separate from the Cluster commands. All OTA Providers SHALL support the [BDX Protocol](#) to allow for the downloading of OTA images by both sleepy End Devices and more capable devices, without requiring access to the public Internet from the OTA Requestor. OTA Requestors SHOULD support the [BDX Protocol](#).

In order to maximize the interoperable combinations of deployed products and Fabric Administrators, the Connectivity Standards Alliance’s Distributed Compliance Ledger (DCL) MAY contain sufficient OTA Software Update information to cover a large number of products, using a federated mechanism of data maintenance. See [Section 11.23, “Distributed Compliance Ledger”](#) for details on the Distributed Compliance Ledger common data schemas. See [Section 11.20.3.3.2, “Conceptual algorithm for matching OTA Software Images applicable to a query”](#) for the conceptual algorithm recommended for implementation by OTA Providers to match records available in the DCL to